

# Using Artificial Intelligence-Assisted Programming Tools to Enhance Undergraduate Computer Science Learning



A Quasi-Experimental Study

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Gao Kangjia

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Hebei University



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# 1. Introduction





AI coding assistants such as GitHub Copilot and ChatGPT are rapidly entering university classrooms. **Yet empirical evidence of their educational impact remains limited.**

This study explores how integrating AI-assisted programming tools affects:

- Students' conceptual understanding
- Students' programming performance
- Students' programming self-efficacy



- **Promise:** AI assistants are a proven productivity multiplier in industry. Can we adapt this power for student learning?
- **Peril:** Without guidance, students risk becoming dependent on AI, potentially hindering the development of fundamental skills.
- **Path Forward:** Explicit instruction in critical AI collaboration can transform these tools from a simple crutch into a powerful cognitive scaffold.
- **Imperative:** This educational shift requires a strong evidence base. Our study is designed to provide it.

## 2. Research Questions

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Our research seeks to answer three core questions:

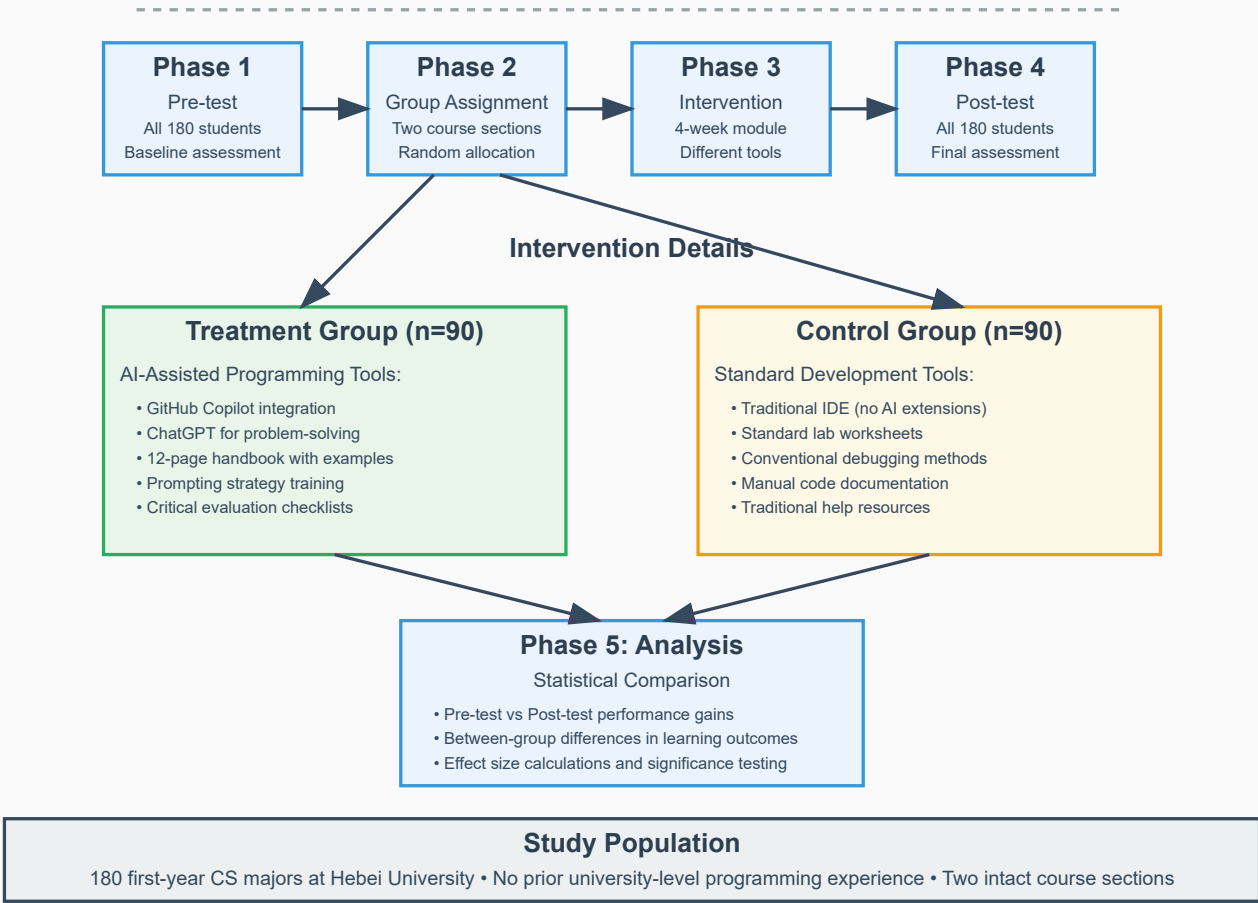
1. **Knowledge:** Does AI assistance improve students' conceptual understanding of programming?
2. **Performance:** Does it enhance the quality of their practical coding assignments?
3. **Confidence:** Does it positively influence their programming self-efficacy?

### **3. Methods**

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Quasi-Experimental Design Timeline





We assess impact across three domains using validated instruments:

- **Conceptual Knowledge:** A 25-item CS1 concept inventory (KR-20 = 0.82).
- **Programming Performance:** A final project graded via a detailed rubric ( $\alpha = 0.87$ ).
- **Self-Efficacy:** A 10-item Likert survey on programming confidence ( $\alpha = 0.91$ ).



- **Power:** A priori analysis confirms our sample ( $n=180$ ) is sufficient to detect a medium effect ( $f=0.25$ ) with  $>95\%$  power.
- **Baseline:** Independent  $t$ -tests will check for pre-existing group differences.
- **Gains:** Matched-sample  $t$ -tests and ANCOVA (controlling for pre-test scores) will evaluate learning gains.
- **Effect Size:** We will report Cohen's  $d$  and partial  $\eta^2$  to quantify the magnitude of effects.

## 4. Expected Contributions

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We anticipate our findings will make three key contributions:

- **For Researchers:** Establishing a rigorous empirical benchmark on the impact of AI tools in introductory programming education.
- **For Educators:** Providing evidence-based guidelines for the responsible integration of AI into the CS curriculum.
- **For the Community:** Supplying a full suite of open educational resources (OER) to support replication and extension studies.

Thank You